

Milestone 4 Report

Security Risk Assessment, Predictive Intelligence & Executive Analytics.

1. Objective

The objective of Milestone 4 was to move the existing incident analytics solution beyond historical reporting toward **security risk assessment, predictive intelligence, and executive-level decision support**. The dashboard focuses on measuring threat exposure, identifying operational vulnerabilities and resource gaps, comparing risk across locations and operational units, forecasting future incident activity, and providing management with consolidated risk indicators for strategic decision-making.

2. Dashboard

Milestone 4 consists of three major dashboards:

Security Risk Assessment Dashboard

This dashboard includes:

- Threat Exposure Level Analysis
- Operational Vulnerability Analysis
- Risk Indicators by Location
- Risk Indicators by Operational Unit
- Resource Allocation Gap Analysis
- Security Weakness Analysis
- Security Risk Scorecard

Predictive Intelligence Dashboard

This dashboard includes:

- Predicted Threat Risk
- Incident Trend Forecast
- Emerging Risk Forecast
- Potential High-Risk Zones
- Forecast Evaluation

Power BI's built-in **time-series forecasting** was used to estimate future incident activity based on historical trends.

Executive Intelligence Dashboard

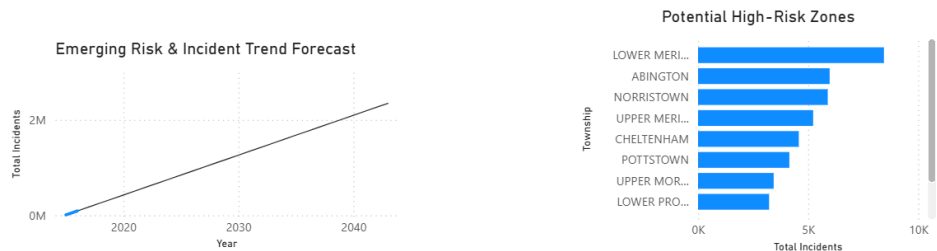
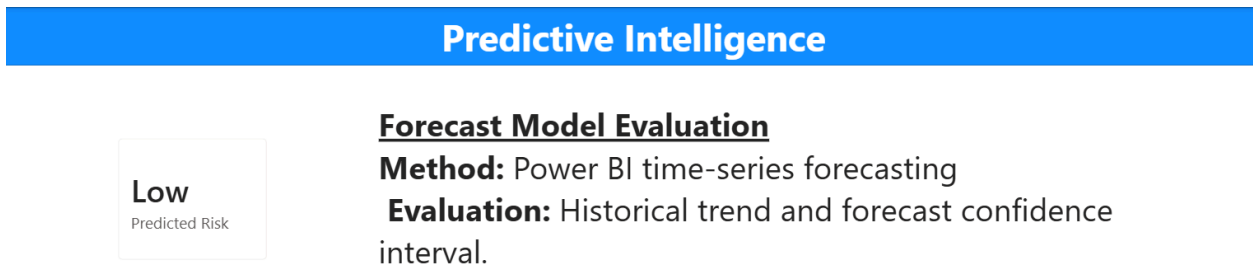
This dashboard consolidates:

- Overall Risk Score
- High-Risk Locations
- Critical Risk Indicators

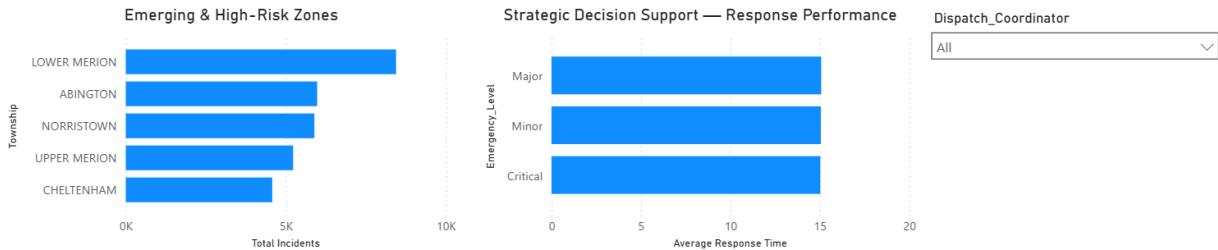
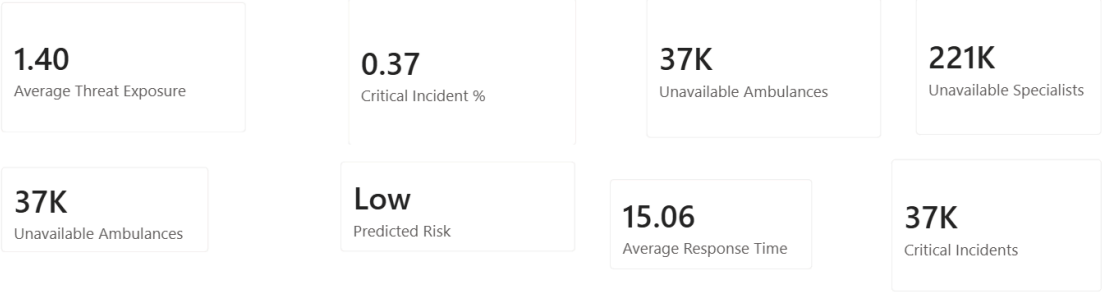
- Predicted Risk
- Emerging Risk Zones
- Operational Risk Indicators
- Strategic Decision-Support Metrics

Interactive slicers were implemented for **Year**, **City**, **Severity**, **Category**, **Operational Unit**, and **Risk Level**.

Dashboard Screenshots:



Executive Intelligence Dashboard



Link: https://drive.google.com/file/d/1zAgLcjDt3QpXCz3zhVck4h8ltBiwokkl/view?usp=drive_link

3. Five Key Insights

Insight 1 — Severity-Weighted Threat Exposure

A severity-weighted risk score was developed by assigning greater weight to higher-severity incidents. This provides a consolidated measure of threat exposure and makes it easier to distinguish between low- and high-risk operational conditions.

Insight 2 — Operational Resource Gaps

The analysis of ambulance and specialist availability highlights situations where required resources may be unavailable. These resource gaps represent potential operational vulnerabilities and can affect emergency response capability.

Insight 3 — Geographical Risk Concentration

Locations with high incident concentration were identified as potential high-risk zones. These areas should receive greater monitoring and preparedness because repeated incident activity indicates higher exposure.

Insight 4 — Response Performance

Response time varies across severity levels, locations, and operational coordinators. This provides management with an indication of where response efficiency may require improvement or additional resources.

Insight 5 — Future Incident Activity

The time-series forecast provides an estimate of future incident activity based on historical patterns. Periods showing an upward forecast can be treated as potential emerging-risk periods and can support proactive resource planning.

4. Predictive Findings

The predictive component uses **Power BI time-series forecasting** rather than a separately trained machine-learning model.

The main findings are:

- Historical incident patterns can be extended into a future forecast to identify potential increases or decreases in incident activity.
- Locations with consistently high historical incident concentration can be treated as **potential high-risk zones** for monitoring and preparedness.
- The rule-based predicted-risk indicator classifies the current threat exposure into **High, Medium, or Low** based on the severity-weighted risk score.
- Forecast confidence intervals provide an indication of uncertainty around the predicted incident trend.

Note: Since no supervised machine-learning model was trained, traditional metrics such as accuracy, precision, recall, and F1-score were not used. This distinction should be retained in the report to avoid overstating the predictive capability.

5. Strategic Recommendations

Recommendation 1 — Prioritize Resources in High-Risk Areas

Management should prioritize emergency resources and monitoring efforts in locations with consistently high incident concentration and elevated risk indicators.

Recommendation 2 — Address Operational Resource Gaps

Frequent resource unavailability, particularly for ambulances and specialist support, should be investigated. Additional capacity or improved resource allocation should be considered where persistent gaps are identified.

Recommendation 3 — Use Forecasts for Proactive Planning

Forecasted increases in incident activity should be incorporated into operational planning. Staffing, emergency resources, and preparedness measures can be adjusted before expected periods of increased demand.